

DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Machine Learning and Deep Learning | Course Code: 21CS71 | Credits: 4

Exam Session: End Semester Exam 2024

Max Marks: 80 | Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

Q1. Derive the mathematical weight update equations for a Multi-Layer Perceptron using the Backpropagation Algorithm with Gradient Descent and Mean Squared Error (MSE) loss function. [8 Marks]

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Q2. Explain the building blocks of Convolutional Neural Networks (CNN): (i) Convolution layer with stride and padding, (ii) ReLU activation, (iii) Max pooling layer, and (iv) Fully Connected classification layer. [8 Marks]